CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

7.5 Polarisation

Syllabus Section: 7. Waves **Paper Type:** Multiple Choice (Paper 1) **Total:** 14 Questions (14 Marks)

Question 1 Easy [1 mark]

The following are statements about waves. Which statement below describes a situation in which polarisation would not happen?

A microwaves pass through a metal grid

B sound waves pass through a metal grid

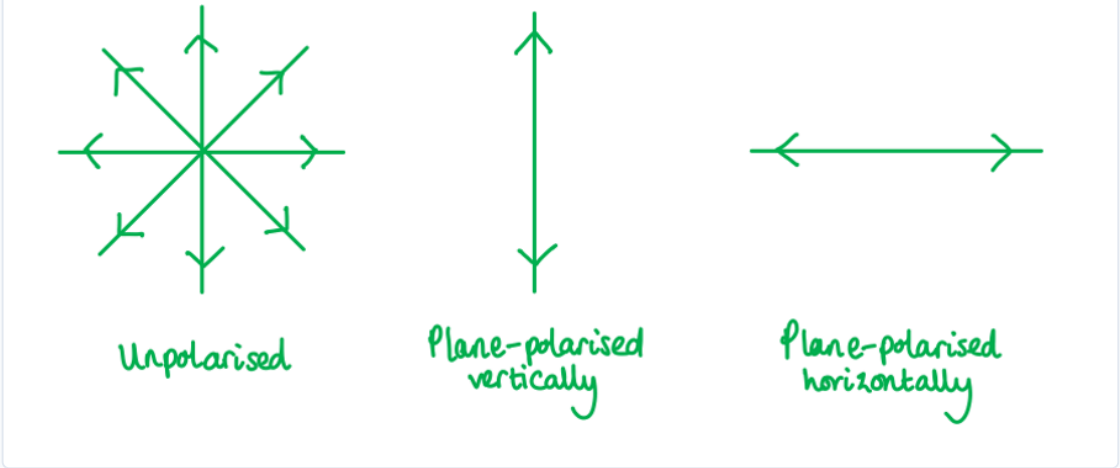
C light waves are reflected

D light waves are scattered

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Polarisation is the process by which oscillations of transverse waves are made to occur in one plane only
- Transverse waves can be polarised as they can oscillate in many planes
- Sound, however, is a longitudinal wave
- Longitudinal waves cannot be polarised as they oscillate parallel to the direction of travel
- Therefore, sound waves would not be polarised by a metal grid



A is incorrect because microwaves are **transverse** waves so can be polarised, this can be done by passing the waves through a grid so that only the waves that fit through the slits can continue

C is incorrect because if light strikes a surface so that there is a 90° angle between the **reflected** and refracted rays, the **reflected** light will be linearly **polarised**

D is incorrect because the scattering of light off air molecules produces linearly polarised light in the plane perpendicular to the incident light

Question 2 Easy [1 mark]

A student looked at the liquid crystal display on their calculator with a polarising film. They observed that as they rotated the film, the display changed.

Which property of the radiation from the calculator display is described correctly?

A the emitted radiation is a transverse wave

B the emitted radiation is a wave with 3 cm wavelength

C the emitted radiation is unpolarised

D the emitted radiation is a longitudinal wave

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Polarisation is the process by which oscillations of transverse waves are made to occur in one plane only
- Transverse waves can be polarised as they can oscillate in many planes
- The radiation emitted from the calculator's display is visible light
- Visible light is part of the electromagnetic spectrum, which is an example of a transverse wave

B is incorrect because a wave with a 3 cm wavelength would be a **microwave**

C is incorrect because as the display is changed as the film rotated this is evidence that the light emitted from the calculator is polarised

D is incorrect because a longitudinal wave can't be polarised as it oscillates parallel to the direction of travel

Question 3 Easy [1 mark]

Which of the following waves can be polarised?

A transverse

B progressive

C stationary

D longitudinal

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Polarisation is the process by which oscillations of transverse waves are made to occur in one plane only
- Transverse waves can be polarised as they can oscillate in many planes
- Light is an example of a transverse wave, and polarisation is a property which is unique to transverse waves only

B is incorrect because the definition of a **progressive** wave is one that moves **through** a material and **transfers energy**

C is incorrect because the definition of a **stationary wave** or **standing wave** is a combination of two waves moving in opposite directions each having the same amplitude and frequency so are in phase. The energy of the two waves cancel each other out. As the wave is not moving through the medium it can't be polarised

D is incorrect because a longitudinal wave can't be polarised as it **oscillates parallel** to the direction of travel

Question 4 Easy [1 mark]

Malus's law shows the intensity of polarised light as it emerges through an analyser. Which is the correct equation to calculate the intensity?

A $I_0 \sin^2 \theta$

B $I_0 \cos^2 \theta$

C $I_0 \cos \theta$

D $\frac{I_0}{2} \cos^2 \theta$

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Polarisation is the process by which the oscillations are made to occur in one plane only
- Malus's law states that the transmitted intensity of the plane polarised light passing through a rotatable polariser is directly proportional to the square of the cosine of the angle through which the polariser is rotated from the position which gives the maximum intensity.
- Malus's law allows us to calculate the intensity of a beam of plane polarised light, it is given by
    - $I = I_0 \cos^2 \theta$

Question 5 Easy [1 mark]

A polarised light beam emerges from a polariser. It is then transmitted onto an analyser, which is at angle ? to the polariser.

What does the value of the angle ? need to be, so that the light emerging from the analyser is at maximum intensity?

A 45°

B 90°

C 180°

D 270°

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- An angle of 180° means that the analyser is parallel to the light transmitted from the polariser
- All of the light incident upon the analyser will emerge from the analyser
    - Therefore the light will be at its maximum intensity

Option **B** and **D** are incorrect because

- Angles of 90° and 270° means that the analyser is perpendicular to the light transmitted from the polariser
- No light will emerge from the analyser at these angles

Option **A** is incorrect because

- An angle of 45° will mean that some light will emerge from the analyser but it will not be at maximum intensity
- In an exam if you can't remember which angles result in maximum intensity use Malus's Law
    - $I = I_0 \quad cos^2(\theta)$
- When θ is 90° or 270°
    - $I = I_0 \times 0$ = 0
- When is 0° or 180°
    - $I = I_0 \times 1 = I_0$
- When is 45°

- $I = \quad I_0 \times \frac{1}{2} = \frac{I_0}{2}$

Question 6 Medium [1 mark]

A linearly polarised electromagnetic beam has an intensity of 16 W m^{-2} with an angle of incidence of 45° to the vertical.

What is the intensity of the transmitted beam?

A 20 W m^{-2}

B 12 W m^{-2}

C 16 W m^{-2}

D 8 W m^{-2}

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Malus's law allows the calculation of the intensity of the beam of plane polarised light, it is given by
  - $I = I_0 \quad cos^2 \quad \theta$
- The initial intensity is equal to I_0 = 16 W m^{-2}
- The angle of incidence is θ = 45°
  - $I = 16 \times cos^2 \quad 45 = 8\text{W} \quad \text{m}^{-2}$
- Therefore, the intensity of the transmitted beam is 8 W m^{-2}

Question 7 Medium [1 mark]

Unpolarised light of intensity I_0 is incident on a polariser with a vertical transmission axis. The transmitted light is incident on a sheet of material X. After transmission through X the intensity of light is $\frac{I_0}{2}$.

It is suggested that X could be

1	a polariser with a vertical transmission axis
2	a polariser with a horizontal transmission axis
3	non-polarising glass

A 1 & 2

B 2 & 3

C 1 & 3

D 3 only

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Polarisation is the process by which oscillations are made to occur in one plane only
- The half rule states that when light passes through the first polarising filter it will reduce the intensity by $\frac{1}{2}$ so the intensity drop of $\frac{I_0}{2}$ will be caused by the first polariser
- If the first polarising sheet is on a vertical plane, then material X could be either another vertical plane or a non-polarising glass, because neither option will change the polarisation or affect the intensity of the transmitted light

A & B are incorrect because if X was a polariser with a horizontal transmission axis, then the intensity of the transmitted light would be zero

D is incorrect because since the light will be polarised vertically after passing through the first filter, options 1 and 3 are both valid

Question 8 Medium [1 mark]

The quality of the sound produced by a portable radio can be improved by changing the orientation of the aerial.

Which statement will correctly describe the explanation of this improvement?

A the radio waves become polarised as a result of adjusting the aerial

B the radio waves become unpolarised as a result of adjusting the aerial

C the radio waves from the transmitter are unpolarised

D the radio waves from the transmitter are polarised

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Polarisation is the process by which the oscillations are made to occur in one plane only
- Radio waves emitted by an antenna are usually transmitted with a specific polarisation, and therefore, receiving aerials will generally only receive radio waves of this polarisation
- If the aerial is moved and this improves the signal then this proves the radio waves must be polarised

A is incorrect because the waves will not become polarised as a result of adjusting the aerial. The source of the waves will determine whether they are polarised or not.

B is incorrect because the waves will not become unpolarised as a result of adjusting the aerial

C is incorrect because if the waves from the transmitter where unpolarised then an adjustment on the aerial would not make a difference as waves would be travelling in all directions

Question 9 Medium [1 mark]

Unpolarised light of intensity I is incident upon a polariser. The light is then transmitted to a second polariser and then a third. The polarisers have the same orientation.

Which statement is true?

A The intensity of the light emerging from the second polariser is $\frac{1}{4}I$

B The intensity of the light emerging from the second polariser is $2I$

C The intensity of the light emerging from the third polariser is $\frac{1}{8}I$

D The intensity of the light emerging from the third polariser is $\frac{1}{2}I$

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- When unpolarised light is incident on the first polariser, the intensity is reduced by half
  - $I_1 = \frac{1}{2}I$
- When light is incident on further polarisers with the same orientation, the intensity remains the same as the first polariser
- Therefore, the intensity of the third polariser is
  - $I_3 = \frac{1}{2}I$
- This is option **D**

Options **A** and **B** are incorrect because

- The intensity of the second polariser is also the same as the first polariser
  - $I_2 = \frac{1}{2}I$
- When the polarisers are not of the same orientation, Malus's Law should be used
  - $I = I_0 \cos^2 \theta$

Question 10 Medium [1 mark]

Unpolarised light of intensity 8 W m^{-2} is incident upon a polariser. The light is then transmitted to a second polariser at an angle of 25° to that of the first polariser.

What is the intensity of the light beam emerging from the second?

A 1.6 W m^{-2}

B 3.3 W m^{-2}

C 3.6 W m^{-2}

D 6.6 W m^{-2}

Mark Scheme & Explanation:

*The correct answer is **B** because*

- The known quantities are
  - Initial intensity, I_0 = 8 W m^{-2}
  - Angle between the first and second polariser, θ = 25°
- When unpolarised light is incident on the first polariser, the intensity is reduced by half
  - $I_1 = \frac{1}{2} I_0$
  - $I_1 = \frac{1}{2} \times 8 = 4$ W m^{-2}
- There is a 25° angle between the first and second polariser so Malus's Law should be used
  - $I = I_0 \cos^2 \theta$
- As the light incident on the second polariser comes from the first, I_0 will be replaced with I_1
  - $I = I_1 \cos^2 \theta$
  - $I = 4 \times \cos^2 25 = 3.3$ W m^{-2} (2 s.f)
- This is option **B**
- You can't write $\cos^2 \theta$ on your calculator
- In order to to do Malus's Law calculations you can write instead $\cos^2 \theta$ of the following
  - $\cos\theta^2$

- $cos\ \theta \times cos\theta$

Question 11 Hard [1 mark]

A beam of unpolarised light is incident on the first of two parallel polarisers. The transmission axis of the two polarisers are initially parallel.

The first polariser is now rotated about the direction of the incident beam by an angle smaller than 90°.

Which row in the table describes the changes to the intensity and polarisation of the light transmitted through both polarisers?

	Intensity	Polarisation
A	no change	no change
B	no change	different
C	different	different
D	different	no change

A

B

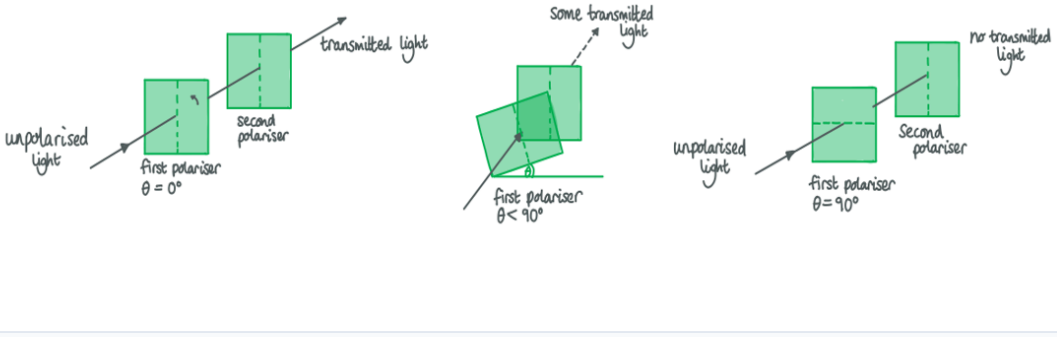
C

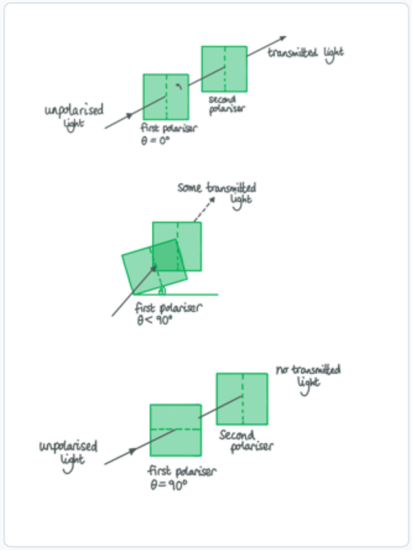
D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

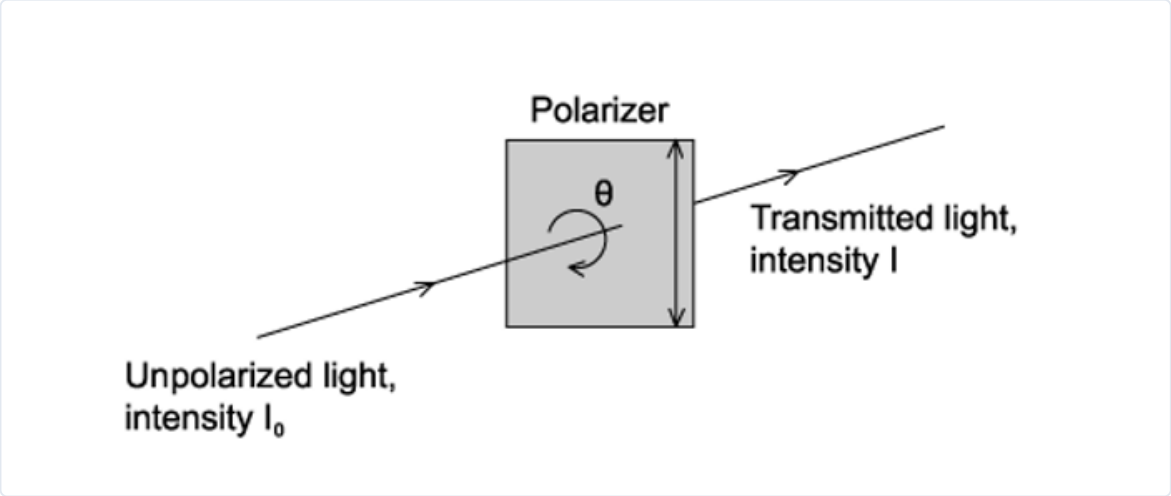
- Polarisation is the process by which the oscillations are made to occur in one plane only
- The first polariser causes the polarisation of the light to be orientated by angle θ
- The first polariser sets the polarisation of the wave and the second only lets through light of that polarisation
- If the second polariser was aligned at 90° to the first, it would block the light completely
- However, since the second polariser is aligned at an angle smaller than 90° to the first, it will allow some light of polarisation with angle θ through and hence reduce the intensity of the light
- It is not possible to polarise already polarised light, so the polarisation does not change


unpolarised light
first polariser
θ = 0°
second polariser
transmitted light
some transmitted light
first polariser
θ < 90°
θ
unpolarised light
first polariser
θ = 90°
Second polariser
no transmitted light

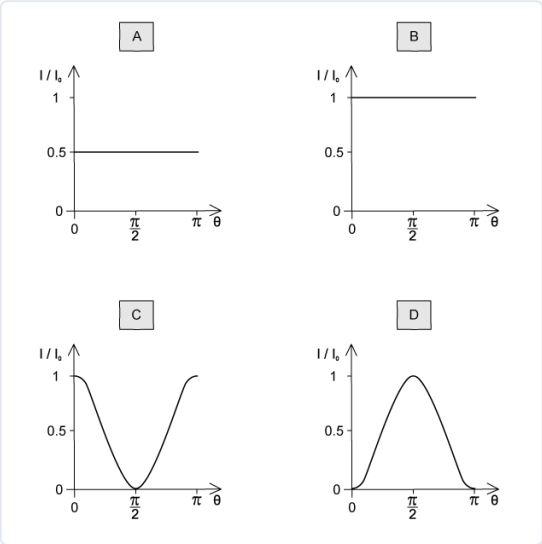

unpolarised light
first polariser
θ = 0°
second polariser
transmitted light
some transmitted light
first polariser
θ < 90°
unpolarised light
first polariser
θ = 90°
Second polariser
no transmitted light

Question 12 Hard [1 mark]

Unpolarised light of intensity I_0 is incident on a polariser that has a vertical transmission axis, as shown in the diagram below.



Which graph shows the intensity of the transmitted light ?



A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Polarisation is the process by which the oscillations are made to occur in one plane only

- As the light passes through the first polariser half of the intensity $\frac{I_0}{2}$ of the ray is lost
    - This is known as the half rule

The other graphs do not recognise that the incoming light is unpolarised, so the other graphs are incorrect

Question 13 Hard [1 mark]

Unpolarised light is incident on a polariser. The light transmitted by the first polariser is then incident on a second polariser. The polarising axis of the second sheet is 60˚ to that of the first.

The intensity emerging from the second polariser is I_2.

Which of the following gives the intensity incident on the first polariser?

A $\frac{I_2}{4}$

B $\frac{I_2}{8}$

C $8I_2$

D $4I_2$

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- Polarisation is the process by which the oscillations are made to occur in one plane only
- As the light passes through the first polariser half of the intensity $\frac{I_0}{2}$ of the ray is lost
    - This is known as the half rule
- Malus's law allows us to calculate the intensity of a beam of plane polarised light, it is given by
    - $I = I_0 \quad cos^2\theta$
- The intensity of light on the second polariser is equal to $= I_0 cos^2 60 = \frac{I_0}{4}$
- Combining the intensity drops from both polarisers, the total intensity of the light after it leaves the second polariser is $\frac{1}{2} \times \frac{I_0}{4} = \frac{I_0}{8}$
    - Therefore, the intensity incident on the first polariser is $8I_2$

Question 14 Hard [1 mark]

Unpolarised light of intensity I_0 is incident on the first of two polarising sheets. Initially the planes of polarisation of the sheets are perpendicular.

Which sheet must be rotated and by what angle θ, so that light intensity $\frac{I_0}{4}$ can emerge from the second sheet ?

	Rotated sheet	Angle θ of rotation
A	1 or 2	$\cos^{-1}\frac{1}{2}$
B	1 or 2	$\cos^{-1}\frac{\sqrt{2}}{2}$
C	2 only	$\cos^{-1}\ \frac{1}{2}$
D	1 only	$\cos^{-1}\frac{\sqrt{2}}{2}$

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Polarisation is the process by which the oscillations are made to occur in one plane only
- As the light passes through the first sheet half of the intensity of the ray is lost, this is known as the half rule
- The second filter will need to reduce the intensity by another half in order for the emerging light to be $\frac{I_0}{4}$
  - So $\cos^2\theta = \frac{1}{2}$
- Rearrange to obtain a value for θ
  - $\cos\theta = \sqrt{\frac{1}{2}}$
  - $\theta = \cos^{-1}\sqrt{\frac{1}{2}} = \cos^{-1}\frac{\sqrt{2}}{2}$
- Either filter could be moved as both will reduce the intensity by half each time the light passes through

A is incorrect because $cos^{-1}\ \frac{1}{2}$ would give an angle of 60˚. This would reduce the intensity of the light greater than 50%

C & **D** are incorrect because either filter could be rotated